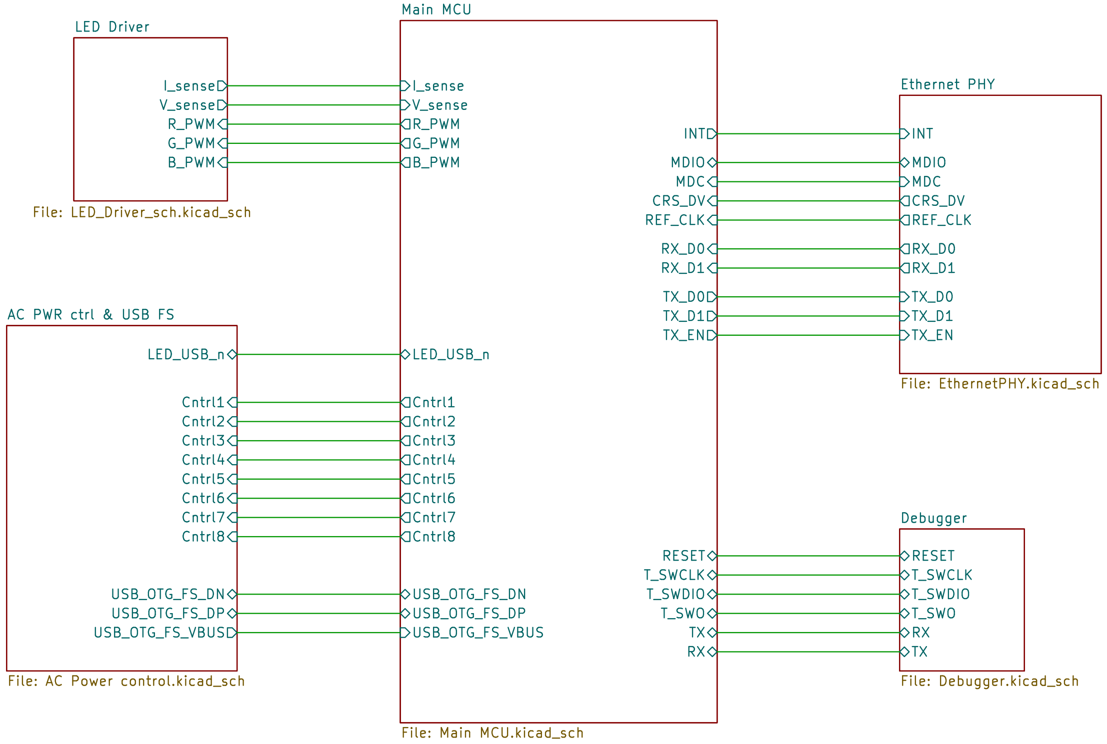


For reference only

Block Diagram

Power Budget

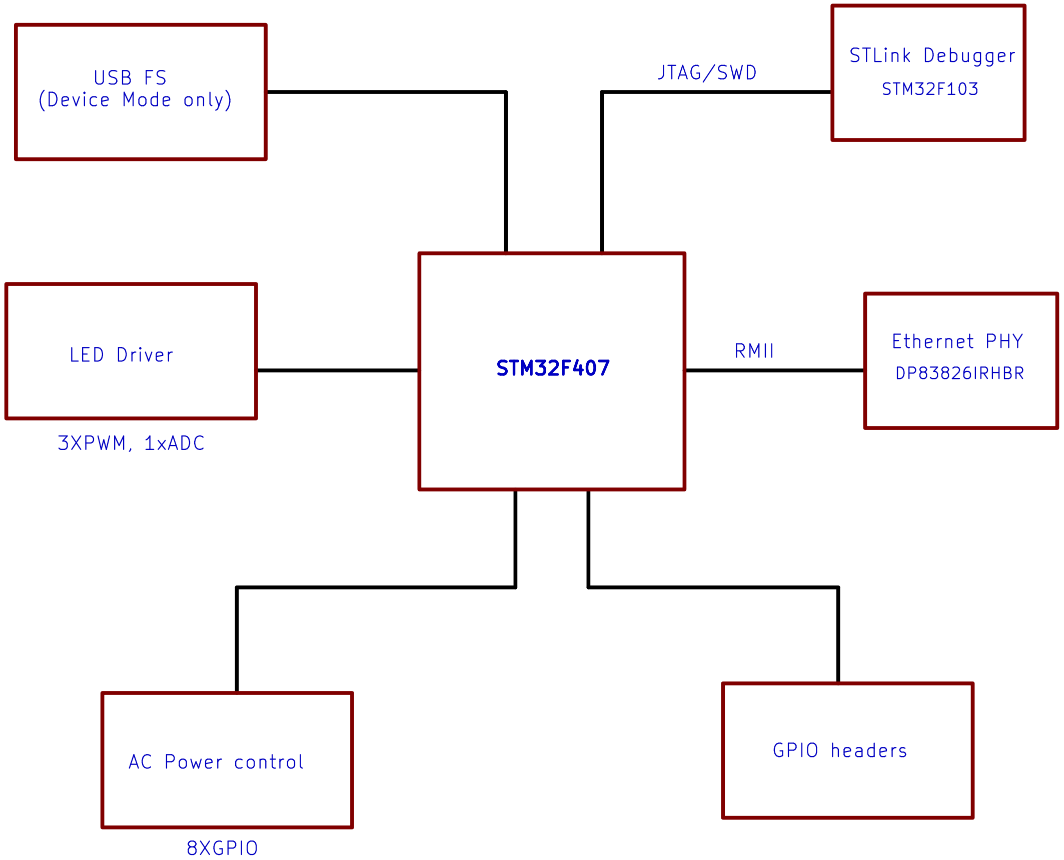
Sheet: /

File: STM32 Microcontroller Board.kicad_sch

Title: DeskNode-F4

Size: A4 | Date: | **Rev: A**

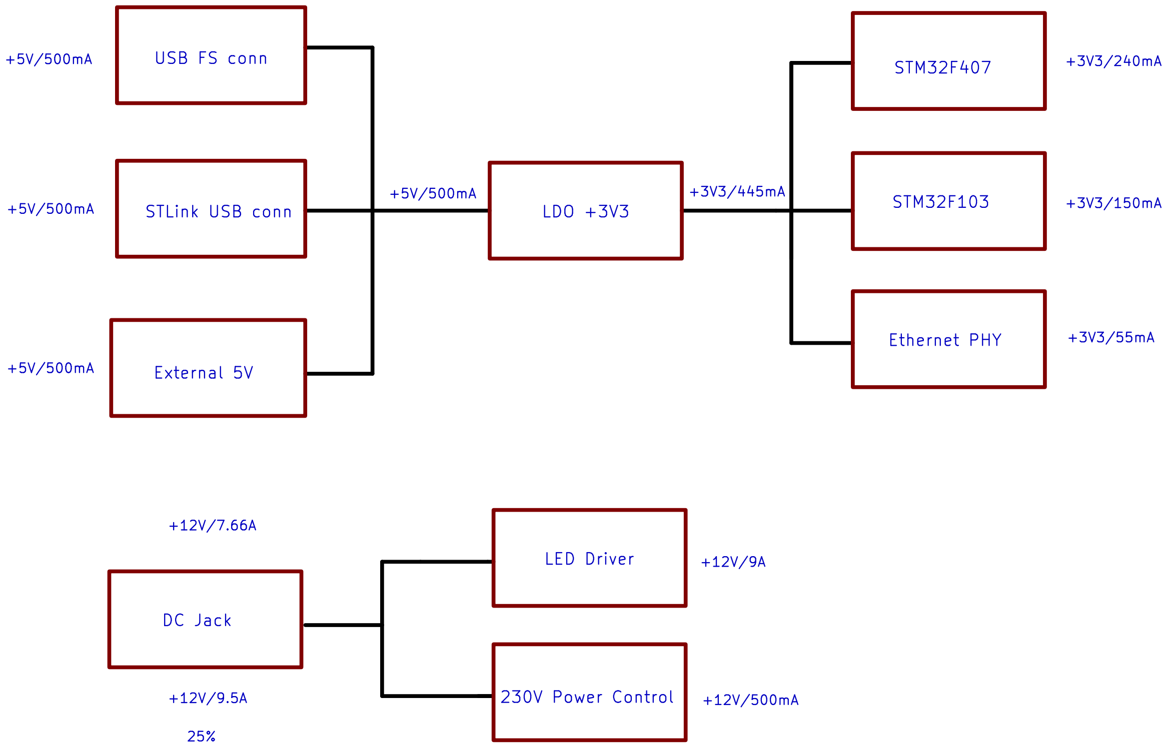
KiCad E.D.A. 9.0.6 | Id: 1/8



Sheet: /Block Diagram/
File: Block Diagram.kicad_sch

Title: DeskNode-F4

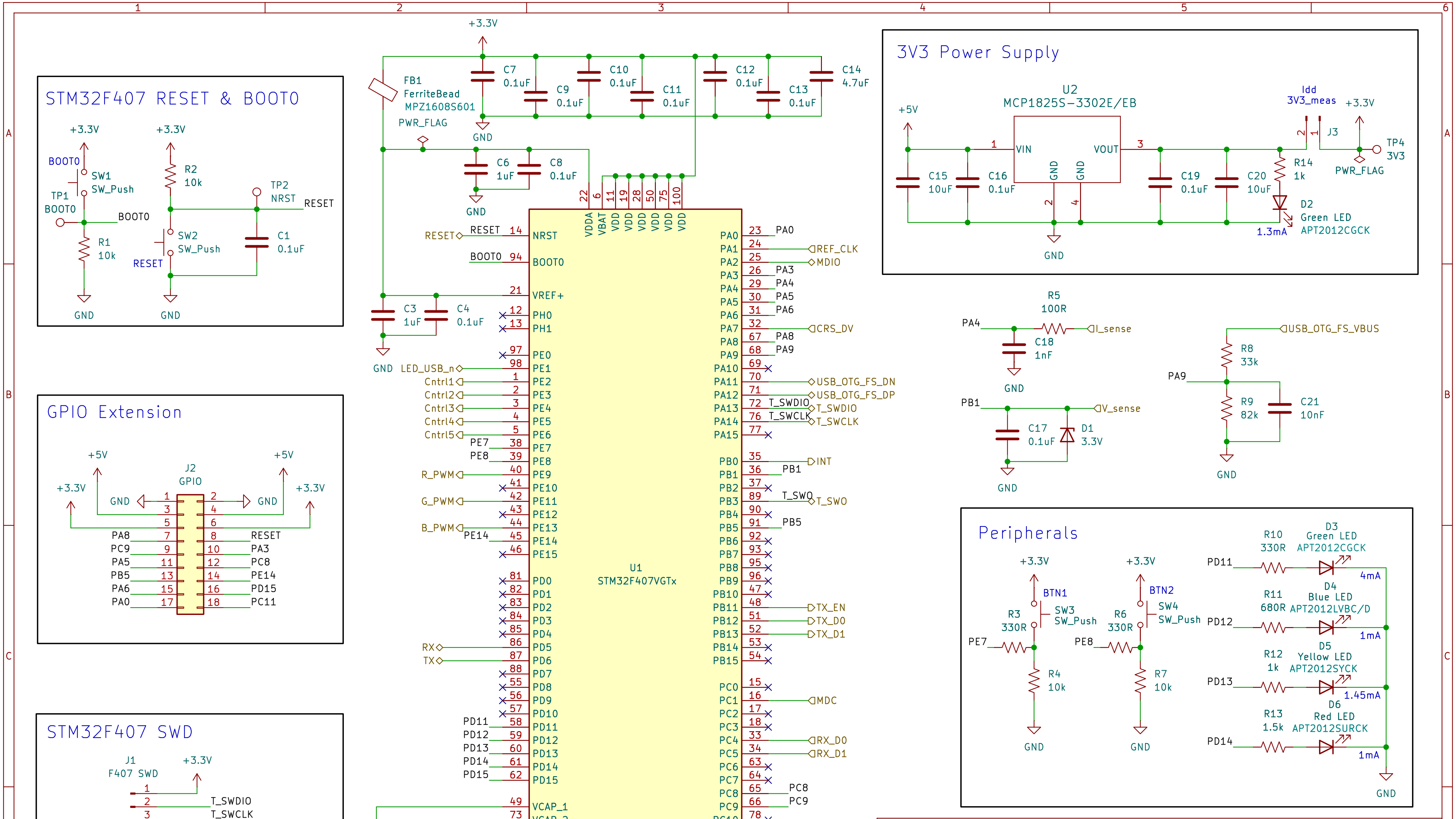
Size: A4	Date:	**Rev: A**
KiCad E.D.A. 9.0.6		Id: 2/8



Sheet: /Power Budget/
File: Power Tree.kicad_sch

Title: DeskNode-F4

Size: A4	Date:	**Rev: A**
KiCad E.D.A. 9.0.6		Id: 3/8

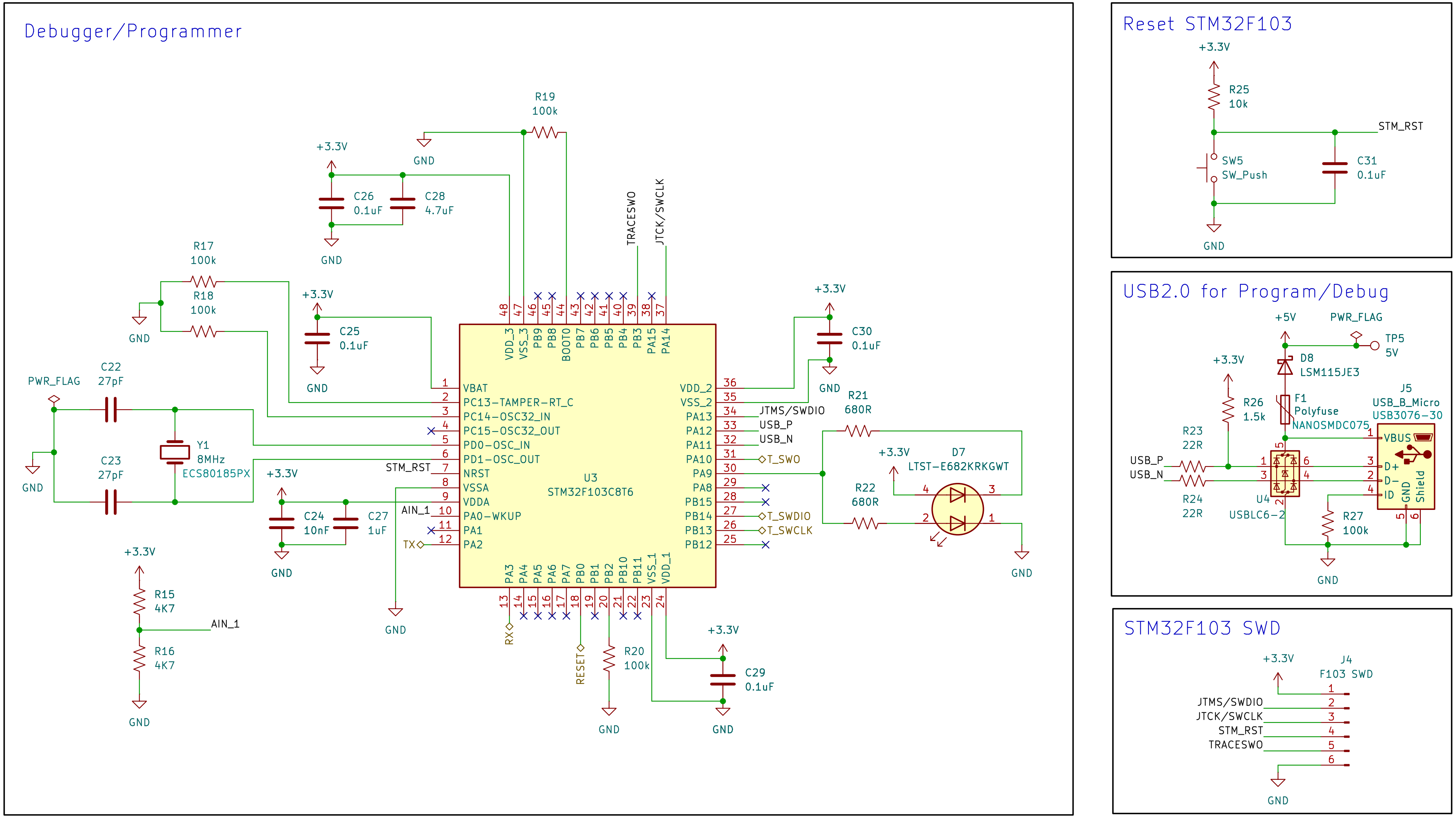


Sheet: /Main MCU/
File: Main MCU.kicad_sch

Title: DeskNode-F4

Size: A4 | Date: | Rev: A

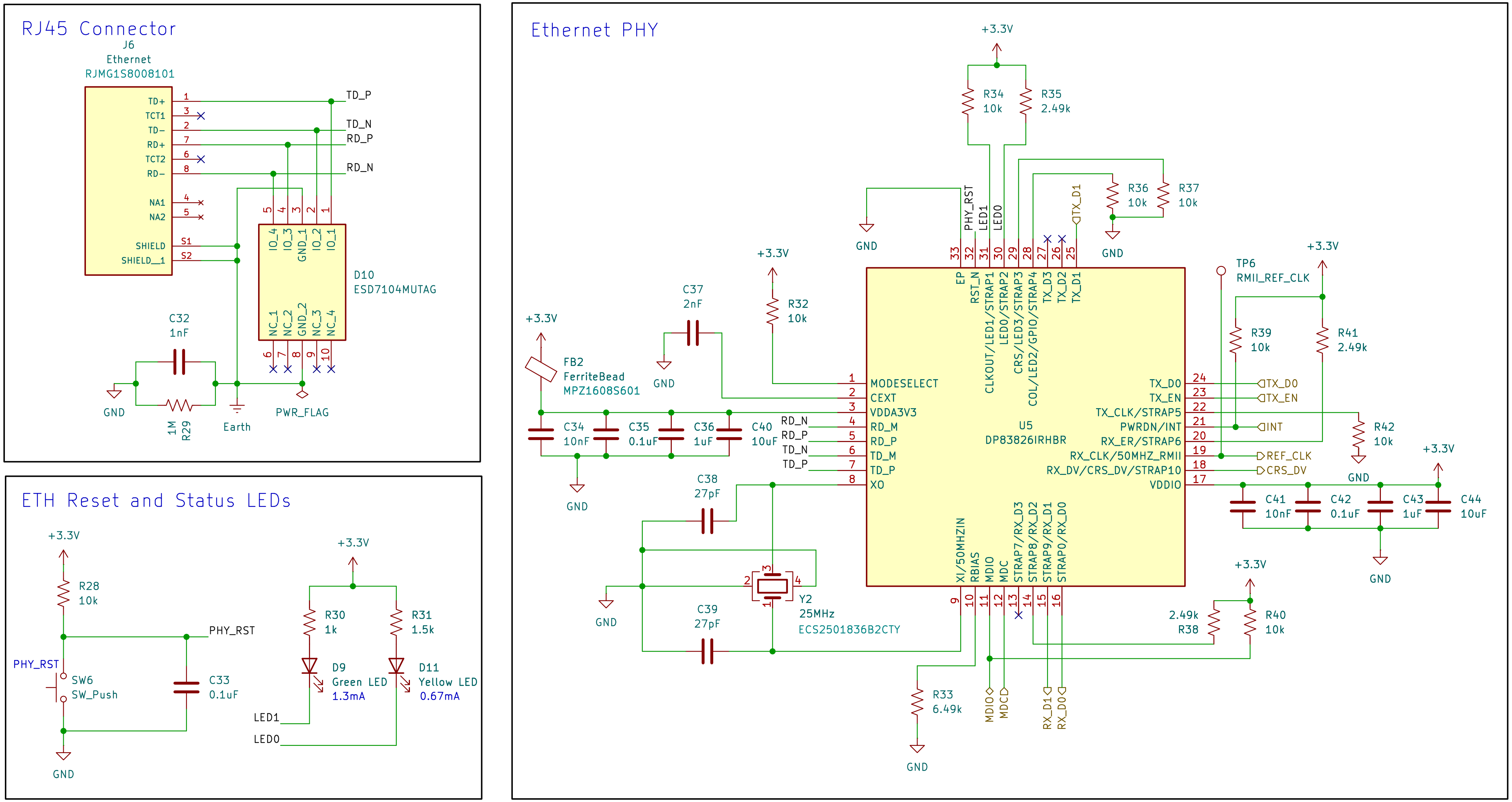
KiCad E.D.A. 9.0.6 | Id: 4/8



Sheet: /Debugger/
File: Debugger.kicad_sch

Title: DeskNode-F4

Size: A4 Date: Rev: A
KiCad E.D.A. 9.0.6 Id: 5/8



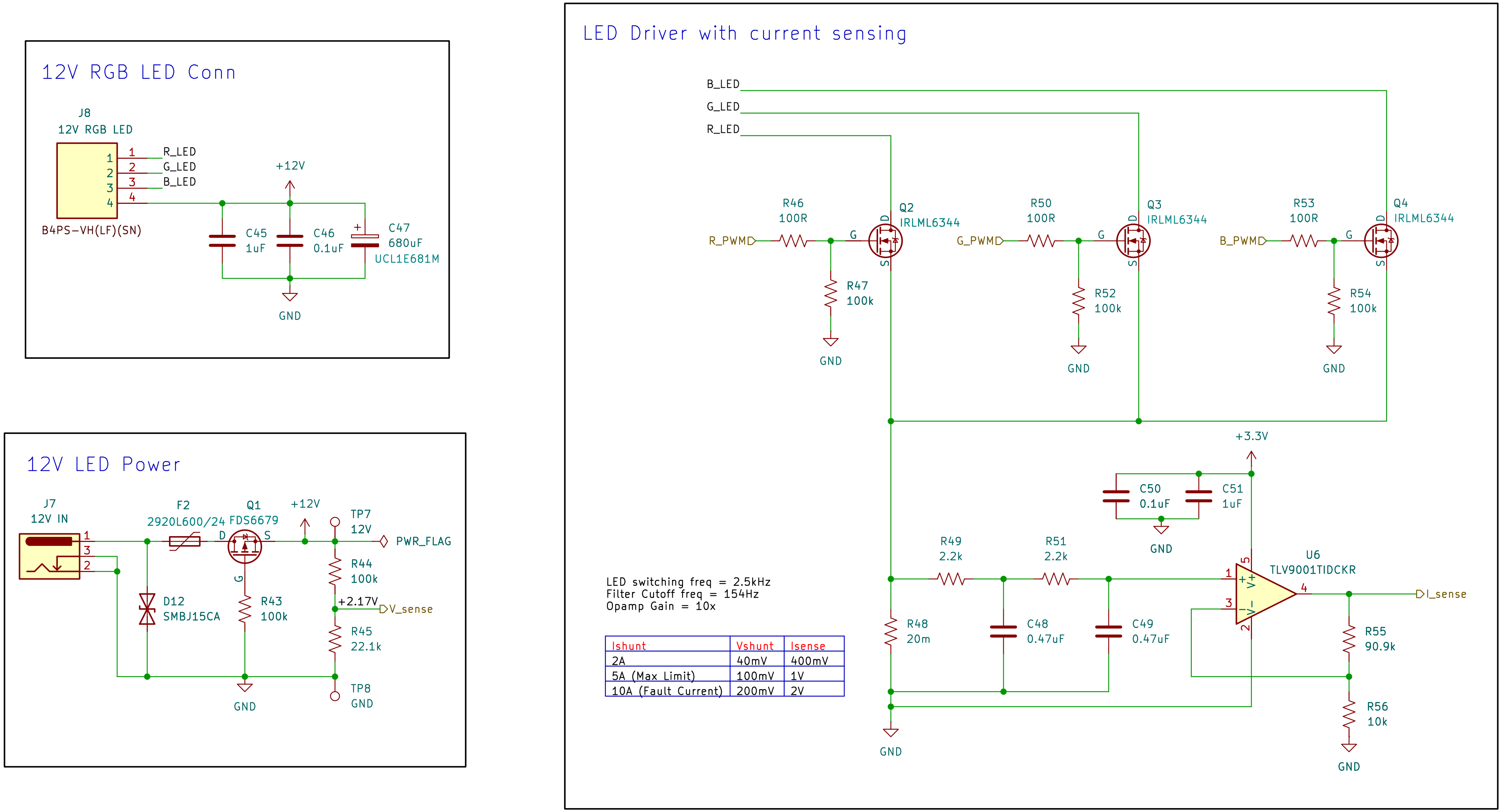
PHY Bootstrap table

Pin Name	Strap	Pin No	Strap Fn	Default	Resistor	Final State
RX_D0	Strap0	16	Auto-negotiation en/dis	0	-	0 - enabled
LED1	Strap1	31	Old Nibble en/dis	1	10k	1 - enabled
LED0	Strap2	30	PHY_ADD0	0	2.49k	1
LED3	Strap3	29	PHY_ADD1	0	10k	0
LED2	Strap4	28	PHY_ADD2	0	10k	0
TX_CLK	Strap5	22	RMII Leader/Follower	0	10k	0 - Leader
RX_ER	Strap6	20	Pin 31 is CLKOUT 25MHz/LED1	0	2.49k	1 - LED1
RX_D3	Strap7	13	Pin 18 is CRS_DV/RX_DV	0	-	0 - CRS_DV
RX_D2	Strap8	14	MII/RMII	0	2.49k	1 - RMII
RX_D1	Strap9	15	auto MDIX en/dis	0	-	0 - enabled
RX_DV	Strap10	18	MDIX/MDI (only applicable if auto-MDIX is dis)	0	-	0 - NA

Sheet: /Ethernet PHY/
File: EthernetPHY.kicad_sch

Title: DeskNode-F4

Size: A4 | Date: | Rev: A
KiCad E.D.A. 9.0.6 | Id: 6/8



LED switching freq = 2.5kHz
Filter Cutoff freq = 154Hz
Opamp Gain = 10x

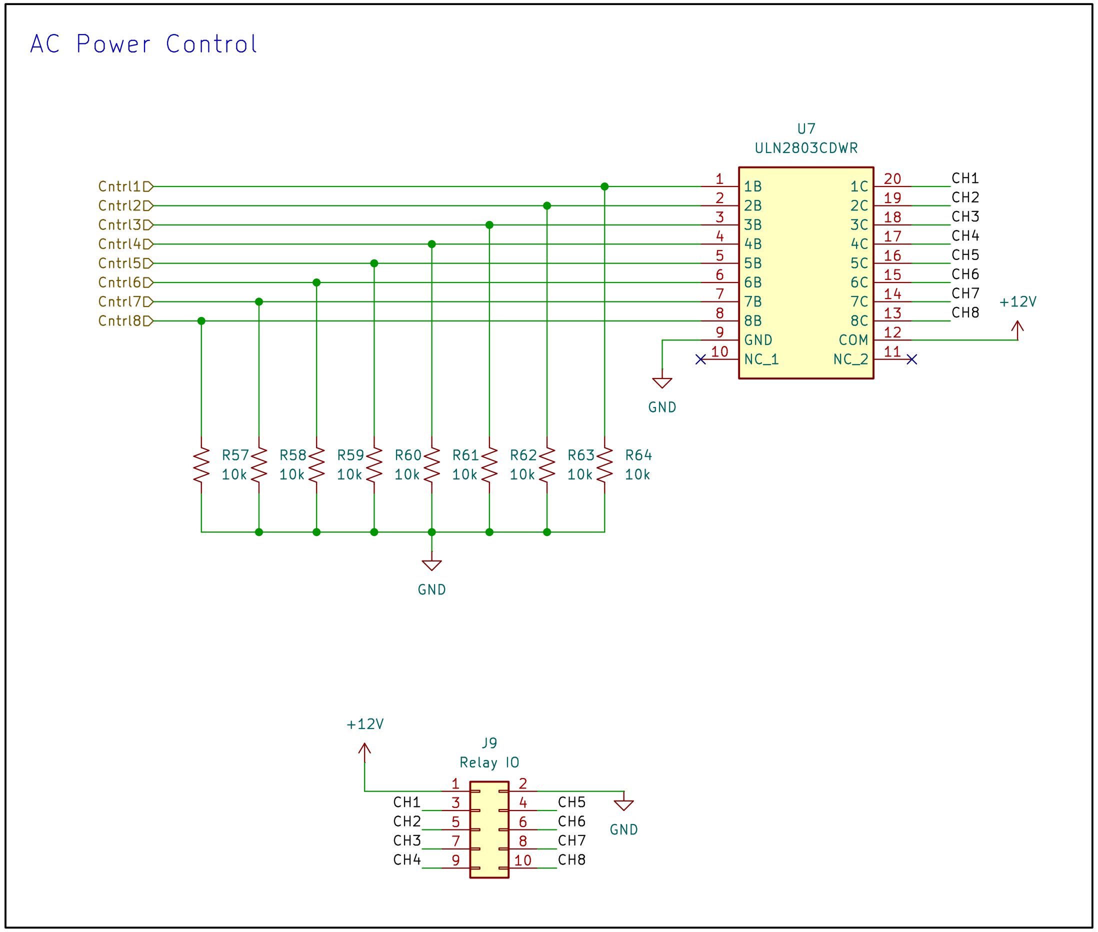
Ishunt	Vshunt	Isense
2A	40mV	400mV
5A (Max Limit)	100mV	1V
10A (Fault Current)	200mV	2V

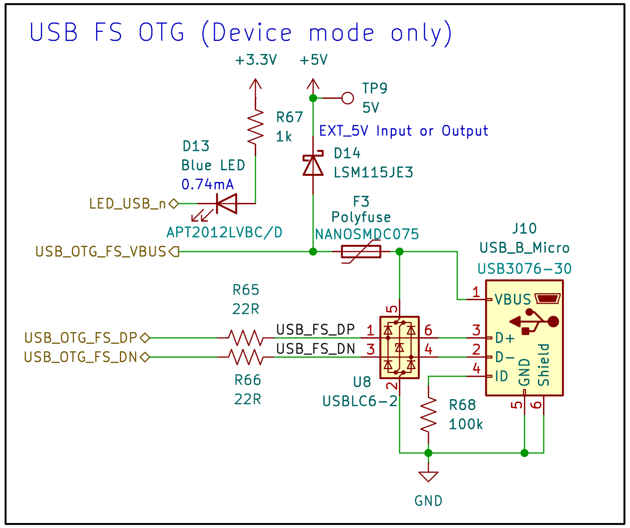
Sheet: /LED Driver/
File: LED_Driver_sch.kicad_sch

Title: DeskNode-F4

Size: A4 | Date: | **Rev: A**

KiCad E.D.A. 9.0.6 | Id: 7/8





Sheet: /AC PWR ctrl & USB FS/
File: AC Power control.kicad_sch

Title: DeskNode-F4

Size: A4	Date:	**Rev: A**
KiCad E.D.A. 9.0.6		Id: 8/8